University of Ottawa

Department of Mathematics and Statistics

MAT 1302C: Mathematical Methods II

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Final Exam

December 7, 2013

Surname _____ First Name _____

Student # _____

Instructions:

(a) You have 3 hours to complete this exam.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1–14	15	16	17	18	19	20	21	22	Total
Maximum	28	4	2	3	4	6	6	4	5	62
Grade										

Part A: Answer Only Questions

In Questions 1–14, only your final answer will be considered for marks. Each question is worth 2 points.

1. Let $A = \begin{bmatrix} -1 & 0 \\ -\frac{1}{2} & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & \frac{3}{2} \\ -1 & 0 \\ 2 & 1 \end{bmatrix}$. Compute AB^T.

Solution: $AB^T = \begin{bmatrix} -2 & 1 & -2 \\ 2 & \frac{1}{2} & 1 \end{bmatrix}$

2. Let $z = 3i - 1$ and $w = 3 + i$. Write the complex number $\dfrac{z + \overline{w}}{w}$ in the form $a + bi$ where a and b are real numbers.

Solution:

$$\frac{z+\overline{w}}{w} = \frac{3i-1+3-i}{3+i} = \frac{2i+2}{3+i} = \frac{(2i+2)(3-i)}{(3+i)(3-i)} = \frac{4i+8}{10} = \frac{8}{10} + \frac{4}{10}i = \frac{4}{5} + \frac{2}{5}i.$$

3. Let $A = \begin{bmatrix} 3 & -2 & -2 & 4 \\ 0 & -2 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix}$. Write down $\operatorname{rank}(A)$ and the dimension of $\operatorname{Nul} A$.

Solution: $\operatorname{rank} A = 3$ and $\dim \operatorname{Nul} A = 1$.

4. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 3 & 0 \\ 0 & 1 & -\frac{1}{2} \end{bmatrix}$. Write down the eigenvalues of A^3 and A^{-1}.

Solution:
Eigenvalues of A^3: 1, 27, $-\frac{1}{8}$.

Eigenvalues of A^{-1}: 1, $\frac{1}{3}$, -2.

5. Let A be an invertible $n \times n$ matrix. Which of the following statements are false?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) The matrix equation $A\mathbf{x} = \mathbf{0}$ is consistent.
(b) The scalar $\lambda = 0$ is an eigenvalue of A.
(c) $\operatorname{rank}(A) = n$.
(d) The matrix A^2A^T is invertible.
(e) A is necessarily diagonalizable.

Solution: (b), (e)

6. Let A, B, and C be 3×3 matrices such that $\det(A) = -\frac{1}{2}$, $\det(B) = \frac{1}{3}$, and $\det(C) = 4$. Calculate $\det(-A^T B^{-1} A^2 C A^{-1})$.

Solution: $\det(-A^T B^{-1} A^2 C A^{-1}) = (-1)^3(-\frac{1}{2})(3)(-\frac{1}{2})^2(4)(-2) = -3$.

7. Let $A = \begin{bmatrix} i & i+1 & \frac{i}{i+2} & 3 \\ 0 & -\frac{1}{i} & \frac{1}{2}+i & 2 \\ 0 & 0 & -i^2 & -2i+3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. Write down the eigenvalues of A and their multiplicities.

Solution: The eigenvalues are i with multiplicity 2 and 1 with multiplicity 2.

8. Let A be an $m \times n$ matrix and $\mathbf{b}$ be a vector in $\mathbb{R}^m$. Which of the following statements implies that the matrix equation $A\mathbf{x} = \mathbf{b}$ is consistent?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) Every row of A contains a pivot position.
(b) $\operatorname{rank} A = n$.
(c) $m = n$ and A is invertible.
(d) $m < n$.
(e) $m = n$.

Solution: (a), (c)

9. For each of the following subsets of $\mathbb{R}^3$, determine if it is a subspace or not.
If the given set is a subspace, you should write **yes**. if it is not a subspace, you should write **no**.

(a) $H = \left\{ \begin{bmatrix} a+b \\ b+c \\ c-a \end{bmatrix} \;\middle|\; a, b, c \in \mathbb{R} \right\}$

(b) $H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; x, y, z \in \mathbb{R} \text{ and } x + y \geq 0 \right\}$

(c) $H = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} \right\}$

(d) $H = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \;\middle|\; x_1, x_2, x_3 \in \mathbb{R} \text{ and } x_1 = x_2 + x_3 \right\}$

Solution: (a) and (d) are subspaces, (b) and (c) are not subspaces.

10. Write down the characteristic polynomial of the matrix $A = \begin{bmatrix} 1 & 1 & -5 & 4 \\ 0 & -2 & 3 & 3 \\ 0 & 0 & 2 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Solution: $(1-\lambda)(-2-\lambda)(2-\lambda)(0-\lambda) = (\lambda^3 - 4\lambda)(\lambda - 1) = \lambda^4 - \lambda^3 - 4\lambda^2 + 4\lambda$.

11. Let A be a 10×8 matrix and $\mathbf{b}$ be a vector in $\mathbb{R}^{10}$ such that the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution. Which of the following statements cannot be correct?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) $\operatorname{rank} A = 10$.

(b) The columns of A are linearly independent.
(c) Every row of A contains a pivot position.
(d) $\dim \operatorname{Nul} A = 2$
(e) The equation $A\mathbf{x} = \mathbf{0}$ is consistent.

Solution: (a), (c), (d).

12. Suppose that $\begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix} = -2$. Calculate the following two determinants.

$\begin{vmatrix} a_{2,1} & a_{2,2} & a_{2,3} \\ a_{1,1} & a_{1,2} & a_{1,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix} =$ **Solution:** 2

$\begin{vmatrix} a_{1,1} & 2a_{1,2} & a_{1,3} - 2a_{1,1} \\ a_{2,1} & 2a_{2,2} & a_{2,3} - 2a_{2,1} \\ a_{3,1} & 2a_{3,2} & a_{3,3} - 2a_{3,1} \end{vmatrix} =$ **Solution:** -4

13. Let $A = PBP^{-1}$ where $P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. Compute A^5.

Solution: $A^5 = PB^5P^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

14. Determine <u>all</u> values of the parameter p for which the following linear system is <u>inconsistent</u>.

$$\begin{cases} -2x_1 + 3x_2 - x_3 = 2 \\ x_2 + x_3 = -1 \\ -2x_2 + px_3 = -3 \end{cases}$$

Solution: $p = -2$.

Part B: Long Answer Questions

In Questions 15–22, you must show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.

15. **(4 points)** Is the following linear system consistent or inconsistent? If it is consistent then find its general solution and write it in vector parametric form.

$$\begin{cases} x_1 + 5x_2 + 2x_3 = x_4 \\ x_3 + 2 = 2x_4 \\ 3x_1 + 15x_2 + 4x_3 + x_4 = 4 \end{cases}$$

Solution: We find the reduced echelon form of the augmented matrix:

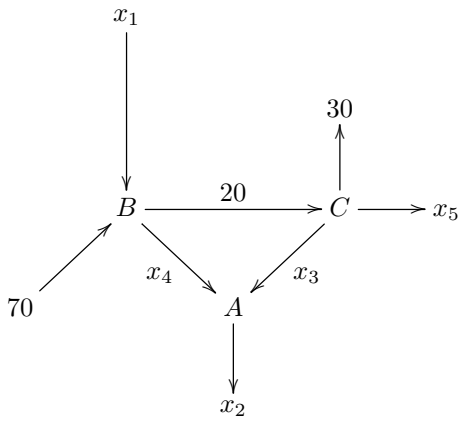
$$\left[\begin{array}{cccc|c} 1 & 5 & 2 & -1 & 0 \\ 0 & 0 & 1 & -2 & -2 \\ 3 & 15 & 4 & 1 & 4 \end{array}\right] \xrightarrow{R_3 \to R_3 - 3R_1} \left[\begin{array}{cccc|c} 1 & 5 & 2 & -1 & 0 \\ 0 & 0 & 1 & -2 & -2 \\ 0 & 0 & -2 & 4 & 4 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{cccc|c} 1 & 5 & 2 & -1 & 0 \\ 0 & 0 & 1 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{cccc|c} 1 & 5 & 0 & 3 & 4 \\ 0 & 0 & 1 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\Rightarrow \begin{cases} x_1 = -5x_2 - 3x_4 + 4 \\ x_2 : \text{free} \\ x_3 = 2x_4 - 2 \\ x_4 : \text{free} \end{cases}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -5x_2 - 3x_4 + 4 \\ x_2 \\ 2x_4 - 2 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -5 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \\ -2 \\ 0 \end{bmatrix}$$

16. **(2 points)** The traffic flow in a city is represented by the diagram below. The arrows indicate the directions of one-way traffic. Numbers and variables indicate the number of cars per hour in the corresponding direction. Write down a linear system describing the traffic flow. Do not solve the linear system.



Solution:

$A:\ x_3 + x_4 = x_2$

$B:\ x_1 + 70 = x_4 + 20$ (or $x_1 + 50 = x_4$)

$C:\ 20 = 30 + x_3 + x_5$ (or $0 = 10 + x_3 + x_5$)

Total: $70 + x_1 = 30 + x_2 + x_5$ (or $40 + x_1 = x_2 + x_5$)

17. **(3 points)** Find <u>all</u> values of the parameter t such that the set of vectors

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 4 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \\ 7 \\ -6 \end{bmatrix}, \begin{bmatrix} 1 \\ -4 \\ t \\ 1 \end{bmatrix} \right\}$$

is a basis for $\mathbb{R}^4$.

Solution:

$$\begin{bmatrix} 2 & -3 & 5 & 1 \\ 0 & 2 & 3 & -4 \\ 0 & 4 & 7 & t \\ -2 & 3 & -6 & 1 \end{bmatrix} \xrightarrow{R_4 \to R_4 + R_1} \begin{bmatrix} 2 & -3 & 5 & 1 \\ 0 & 2 & 3 & -4 \\ 0 & 4 & 7 & t \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \begin{bmatrix} 2 & -3 & 5 & 1 \\ 0 & 2 & 3 & -4 \\ 0 & 0 & 1 & t+8 \\ 0 & 0 & -1 & 2 \end{bmatrix} \xrightarrow{R_4 \to R_4 + R_3} \begin{bmatrix} 2 & -3 & 5 & 1 \\ 0 & 2 & 3 & -4 \\ 0 & 0 & 1 & t+8 \\ 0 & 0 & 0 & t+10 \end{bmatrix} \Rightarrow \ t \neq -10.$$

18. **(4 points)** Let $A = \begin{bmatrix} 1 & -2 & \frac{1}{2} \\ -2 & 4 & s-1 \\ 2 & -4 & 1 \end{bmatrix}$. First determine the value of s such that $\dim \operatorname{Nul} A = 2$. Then find a basis of $\operatorname{Nul} A$ for this value of s.

Solution:

$$\begin{bmatrix} 1 & -2 & \frac{1}{2} \\ -2 & 4 & s-1 \\ 2 & -4 & 1 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 2R_2]{R_2 \to R_2 + 2R_1} \begin{bmatrix} 1 & -2 & \frac{1}{2} \\ 0 & 0 & s \\ 0 & 0 & 0 \end{bmatrix}$$

For $s \neq 0$ the first and third columns are pivot columns and Therefore

$$\dim \operatorname{Nul} A = 3 - \operatorname{rank} A = 3 - 2 = 1.$$

therefore $\dim \operatorname{Nul} A = 2$ exactly when $s = 0$. In this case the general solution is

$$\begin{cases} x_1 = 2x_2 - \frac{1}{2}x_3 \\ x_2 : \text{free} \\ x_3 : \text{free} \end{cases}$$

and in vector parametric form we have

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$$

and a basis for $\operatorname{Nul} A$ is given by

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix} \right\}$$

19. The city of Chicago will never forget the first week of November: every single day, the controversial statements by the city's mayor made it to the morning news headlines. Surveys indicate that, every day, 40% of the mayor's supporters would change their minds and become his opponents, whereas 20% of his opponents would change their minds and become his supporters.

(a) **(1 point)** Write the migration matrix M.

Solution: $\begin{bmatrix} .6 & .2 \\ .4 & .8 \end{bmatrix}$ (Note that another possibility is $\begin{bmatrix} .8 & .4 \\ .2 & .6 \end{bmatrix}$

(b) **(2 points)** Surveys show that Monday morning, before the mayor's first controversial statement, exactly half of Chicago's population were the mayor's supporters, and the remaining half were his opponents. Determine the percentage of his supporters and opponents on Tuesday evening.

Solution: $\mathbf{x}_2 = M^2\mathbf{x}_0 = \begin{bmatrix} .6 & .2 \\ .4 & .8 \end{bmatrix}^2 \begin{bmatrix} .5 \\ .5 \end{bmatrix} = \begin{bmatrix} .44 & .28 \\ .56 & .72 \end{bmatrix} \begin{bmatrix} .5 \\ .5 \end{bmatrix} = \begin{bmatrix} .36 \\ .64 \end{bmatrix}$ Therefore 36% of the population will be his supporters and 64% will be his opponents.

(c) **(3 points)** Assuming that the mayor did not stop making controversial statements on a daily basis, what would be the percentage of his supporters and opponents in the long run?

Solution:

$M\mathbf{x} = \mathbf{x} \Rightarrow (M - I_2)\mathbf{x} = \mathbf{0} \Rightarrow \begin{bmatrix} -.4 & .2 \\ .4 & -.2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\left[\begin{array}{cc|c} -.4 & .2 & 0 \\ .4 & -.2 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + R_1} \left[\begin{array}{cc|c} -.4 & .2 & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to -\frac{10}{4}R_1} \left[\begin{array}{cc|c} 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{cases} x_1 = \frac{1}{2}x_2 \\ x_1 + x_2 = 1 \end{cases}$

$\Rightarrow 1 - x_1 = x_2 = 2x_1 \Rightarrow x_1 = \frac{1}{3}$ and $x_2 = \frac{2}{3}$

Therefore the long term percentages are: $\begin{cases} \text{Supporters: } \frac{1}{3} \simeq 33\% \\ \text{Opponents: } \frac{2}{3} \simeq 66\% \end{cases}$

20. (a) **(2 points)** Let $A = \begin{bmatrix} 3 & 1 & -2 \\ 0 & -1 & 0 \\ -1 & 5 & 4 \end{bmatrix}$. Write the characteristic polynomial of A and compute its eigenvalues.

Solution:
By expanding along the second row we have:

$$\begin{aligned} \det(A - \lambda I_3) &= \begin{vmatrix} 3-\lambda & 1 & -2 \\ 0 & -1-\lambda & 0 \\ -1 & 5 & 4-\lambda \end{vmatrix} = (-1-\lambda)\begin{vmatrix} 3-\lambda & -2 \\ -1 & 4-\lambda \end{vmatrix} \\ &= (-1-\lambda)\left((3-\lambda)(4-\lambda) - (-2)(-1)\right) = -(\lambda+1)(\lambda^2 - 7\lambda + 10) \\ &= -(\lambda+1)(\lambda-2)(\lambda-5) \end{aligned}$$

and therefore the eigenvalues of A are $\lambda = -1$, $\lambda = 2$, $\lambda = 5$.

(b) **(4 points)** Let $B = \begin{bmatrix} 4 & -1 & 1 \\ 2 & 1 & 2 \\ -1 & 1 & 2 \end{bmatrix}$. Find a basis for the eigenspace of B that corresponds to the eigenvalue $\lambda = 3$.

Solution:

$$B - 3I_3 = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -2 & 2 \\ -1 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & -2 & 2 \\ -1 & 1 & -1 \end{bmatrix} \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \Rightarrow \quad x_1 = x_2 - x_3$$

$$\Rightarrow \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 - x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{ basis for the eigenspace of } \lambda = 3 \ : \ \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

21. **(4 points)** Compute the inverse of $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 3 \\ \frac{2}{3} & 1 & 2 \end{bmatrix}$.

Solution:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 0 \\ \frac{2}{3} & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - \frac{2}{3}R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & -1 & 1 & 0 \\ 0 & \frac{1}{3} & 2 & -\frac{2}{3} & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & \frac{1}{3} & 2 & -\frac{2}{3} & 0 & 1 \\ 0 & 0 & 3 & -1 & 1 & 0 \end{array}\right] \xrightarrow{R_3 \to \frac{1}{3}R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & \frac{1}{3} & 2 & -\frac{2}{3} & 0 & 1 \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{1}{3} & 0 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - 2R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & 0 & -\frac{2}{3} & 1 \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{1}{3} & 0 \end{array}\right] \xrightarrow{R_2 \to 3R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -2 & 3 \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{1}{3} & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -3 \\ 0 & 1 & 0 & 0 & -2 & 3 \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{1}{3} & 0 \end{array}\right]$$

Therefore $A^{-1} = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 3 \\ -\frac{1}{3} & \frac{1}{3} & 0 \end{bmatrix}$.

22. An economy has two sectors: Agriculture and Energy. We know that:

- In order to produce one unit of output, the Agriculture sector should consume .3 units from itself and .4 units from the Energy sector.
- In order to produce one unit of output, the Energy sector should consume .7 units from itself and .5 units from the Agriculture sector.

(a) **(1 point)** Write down the consumption matrix for this economy.

Solution: $C = \begin{bmatrix} .3 & .5 \\ .4 & .7 \end{bmatrix}$ (Note that $C = \begin{bmatrix} .7 & .4 \\ .5 & .3 \end{bmatrix}$ is another valid answer.)

(b) **(1 point)** Determine the intermediate demands that are created if Agriculture wants to produce 15 units and Energy wants to produce 20 units.

Solution: $15 \begin{bmatrix} .3 \\ .4 \end{bmatrix} + 20 \begin{bmatrix} .5 \\ .7 \end{bmatrix} = \begin{bmatrix} 14.5 \\ 20 \end{bmatrix}$

(c) **(3 points)** Calculate the production levels that are required to meet a final demand of 5 units from the Agriculture sector and 2 units from the Energy sector.

Solution: $I_2 - C = \begin{bmatrix} .7 & -.5 \\ -.4 & .3 \end{bmatrix} \Rightarrow (I_2 - C)^{-1} = \frac{1}{.01} \begin{bmatrix} .3 & .5 \\ .4 & .7 \end{bmatrix} = \begin{bmatrix} 30 & 50 \\ 40 & 70 \end{bmatrix}$

$\Rightarrow \mathbf{x} = \begin{bmatrix} 30 & 50 \\ 40 & 70 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 250 \\ 340 \end{bmatrix}$

and production levels are $\begin{cases} \text{Agriculture:} & 250 \\ \text{Energy:} & 340. \end{cases}$